Premium Computer Networking Study Guide

Transport Layer • Application Layer • IoT • SDN

Semester Exams • Viva • Interviews • Revision Handbook

Transport Layer

Introduction
The Transport Layer provides end-to-end communication between applications running on hosts. It ensures that packets arriving from the Network Layer are delivered to the correct application.

Functions Segmentation and Reassembly Reliable Delivery Flow Control Error Control Port Addressing Multiplexing and Demultiplexing **Network Example:** A browser communicates with a web server using TCP. The transport layer divides data into segments, numbers them, and reassembles them at the destination.

Quick Revision Box
Key exam facts, definitions and must-remember concepts from this topic.

TCP and UDP

TCP: Reliable, connection-oriented protocol.
3-Way Handshake
1. SYN → Client requests connection.
2. SYN-ACK → Server accepts.
3. ACK → Client confirms.

4-Way Handshake
FIN → ACK → FIN → ACK.

UDP: Connectionless, fast, lightweight, no delivery guarantee.
Used in DNS, streaming, gaming and VoIP.

Quick Revision Box
Key exam facts, definitions and must-remember concepts from this topic.

Port Addressing

Ports identify applications running on a host. HTTP80 HTTPS443 FTP21 SMTP25 DNS53 **Network Analogy:** IP address identifies the building while the port identifies the room.

Quick Revision Box
Key exam facts, definitions and must-remember concepts from this topic.

Congestion Control

Leaky Bucket
Traffic exits at a fixed rate, smoothing bursts.

Token Bucket
Tokens accumulate over time and allow controlled bursts.

Used to prevent routers and network links from becoming overloaded.

Quick Revision Box
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Application Layer Protocols

DNS converts domain names to IP addresses.
HTTP transfers web pages.
HTTPS provides encrypted communication.
FTP transfers files.
SMTP sends email.
POP3 downloads email.

IMAP synchronizes email across devices.

Quick Revision Box
Key exam facts, definitions and must-remember concepts from this topic.

Remote Login & File Sharing

Telnet: Remote login without encryption.
SSH: Secure encrypted remote login.

NFS: Common in Linux/Unix environments.
SMB: Common in Windows environments.

Quick Revision Box
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Domain Name System

DNS is a distributed hierarchical naming system. Process: 1. Browser requests a domain. 2. Resolver checks cache. 3. DNS server lookup occurs. 4. IP address is returned. 5. Browser contacts web server.

Quick Revision Box
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SNMP

Simple Network Management Protocol is used to monitor routers, switches, servers and network devices. Components: • SNMP Manager • SNMP Agent • MIB (Management Information Base)

Quick Revision Box
Key exam facts, definitions and must-remember concepts from this topic.

Cloud Networking

Cloud networking delivers networking services using cloud infrastructure. Features: • Virtual Networks • Load Balancers • Elastic Scaling • Security Groups • High Availability

Quick Revision Box
Key exam facts, definitions and must-remember concepts from this topic.

Internet of Things (IoT)

IoT connects physical devices to networks. Examples: • Smart Watches • Smart Homes • Smart Sensors • Industrial Automation • Connected Vehicles

Quick Revision Box
Key exam facts, definitions and must-remember concepts from this topic.

MQTT and CoAP

MQTT Publish/Subscribe protocol using a broker. Ideal for low-bandwidth environments. **CoAP** Lightweight protocol based on UDP. Designed for constrained IoT devices.

Quick Revision Box
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Software Defined Networking (SDN)

SDN separates: • Control Plane • Data Plane Advantages: • Centralized management • Automation • Faster deployment • Better visibility

Quick Revision Box
Key exam facts, definitions and must-remember concepts from this topic.

Important Comparison Tables

Feature	TCP	UDP
Connection	Connection-Oriented	Connectionless
Reliability	Reliable	Not Reliable
Speed	Slower	Faster
Acknowledgement	Yes	No
Use Cases	Web, Email, FTP	Gaming, Streaming, DNS

Protocol	Purpose	Port
DNS	Name Resolution	53
HTTP	Web Browsing	80
HTTPS	Secure Browsing	443
FTP	File Transfer	21
SMTP	Send Email	25
POP3	Download Email	110
IMAP	Email Sync	143

Exam-Oriented Revision

Frequently Asked Questions Explain TCP 3-Way Handshake. Differentiate TCP and UDP. What is Port Addressing? Explain DNS resolution process. Difference between POP3 and IMAP. Difference between Telnet and SSH. Explain MQTT architecture. What is SDN and why is it important? **Common Confusions** HTTP vs HTTPS POP3 vs IMAP NFS vs SMB MQTT vs CoAP TCP reliability vs UDP speed